Navneeth Krishna Aravind

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EDUCATION

M.Sc. in Applied Data Science and Analytics (GPA: 1.8) Apr 2024 – Mar 2026
SRH Hochschule Heidelberg | Heidelberg, Germany

B.Tech in Mechatronics (GPA: 2.2) Jul 2017 – Jun 2021
Sastra Deemed to Be University | Thanjavur, India

PROJECTS

Improving Forecasting Accuracy in Bike Rental Demand via Advanced Deep Learning Oct 2025 – Mar 2026
- Built hybrid models (LSTM, TCNN, Transformers) for time-series forecasting; achieved R^2 up to **0.92** evaluated via RMSE, MAE, and MAPE.
- Engineered temporal & weather features with XGBoost-based selection to improve model robustness and prediction quality.

Optimizing Deep Learning Models for Text-Based Emotion Detection Jun – Sep 2025
- Developed a BERT + Deep LMU emotion classifier with mean/max/min pooling, achieving **96% validation accuracy** on the CARER dataset.
- Compared data-balancing and pooling techniques to optimize temporal feature representation and overall model performance.

Automated Road Damage Detection May – Jun 2025
- Trained YOLOv5m with Transformer blocks on **8,000 images** across **7 damage types**, achieving **0.814 precision**.
- Benchmarked YOLOv5/v8/v11 across dataset sizes (**1k–6k**) and augmentation strategies to assess robustness and generalization.

Streaming Pulse – Spotify Trend Analysis Apr – May 2025
- Analyzed **10,000 historical** and **100+ real-time** Spotify tracks to uncover audio-feature patterns behind music popularity.
- Built an automated data-processing workflow and Tableau dashboard highlighting factors that distinguish hit songs from average ones.

AI-Powered Metadata Extractor for Biographical Interviews Nov 2024 – Apr 2025
- Built a Streamlit-based LLM app to extract structured metadata from long-form interview transcripts.
- Applied TF-IDF and cosine similarity to improve relevance scoring and ensure metadata consistency.

Tableau's IronViz 2025 – TV Series Storytelling Analysis Jan – Feb 2025
- Analyzed an IMDb dataset of **199,469 TV series** to derive storytelling insights and selected one series for deep-dive analysis.
- Designed an interactive Tableau dashboard with narrative-driven visualizations on Amazon Studios' competitive superhero franchise growth.

YouTube Data Pipeline Nov – Dec 2024
- Built a cloud-based pipeline using Google Cloud, Apache Beam, and YouTube APIs to process **100+ videos** from Sidemen's channels.
- Designed parallel channel- and video-level pipelines feeding Power BI dashboards to reveal viewership patterns and growth opportunities.

Automated Candidate Selection May – Aug 2024
- Analyzed **700+ GitHub profiles**, **15,000+ repositories**, and **1,400+ StackOverflow profiles** using PCA and K-Means clustering.
- Reduced manual screening time by **30%** and improved shortlisting consistency through feature-based candidate ranking.

Food Waste and Environmental Emissions Dashboard Jul – Aug 2024
- Pre-processed FAOSTAT datasets for **100+ countries** (**6,000+ records**) — standardizing types, removing duplicates, creating reliable keys.
- Combined food loss percentages with greenhouse gas emissions (CO_2, CH_4) at commodity and country levels into interactive dashboards.

Big Data Analytics for Strategic Film Production Jun – Jul 2024
- Scraped metadata and images for the **top 100 movies** from **3 rating platforms** and integrated them into MySQL and Hadoop HDFS.
- Enabled large-scale film analytics through distributed storage and structured reporting pipelines.

Perception-Based Navigation of Wheelchair for Entraining & Detraining Feb – May 2021
- Developed an autonomous wheelchair system using YOLOv4 on **9,000+ images**, achieving **97.16% compartment** and **87.46% door detection**.
- Designed and simulated path-planning and navigation logic in VREP using Lua and MATLAB for railway boarding assistance.

Human Behaviour Detection During Online Exams Oct – Dec 2020
- Trained real-time face/gaze detection (OpenCV DNN) and object detection (YOLOv3 + COCO) models to classify suspicious behaviour.

- Integrated models to auto-flag **3 behaviour types** (phone, face absence, multi-person) — enabling reliable automated exam monitoring.

EXPERIENCE

Tata Consultancy Services – System Engineer Oct 2021 – Feb 2024

Tech Stack: Xamarin, MAUI, C#

- Developed and delivered **10+ new features** in an **Agile** environment, using **Jira** for sprint planning and backlog management and **Confluence** for documenting technical specifications and feature walkthroughs, improving user interaction and application capabilities improving user interaction and application capabilities.
- Refactored legacy modules to reduce crashes, improve load times by **30%**, and **create a maintainable codebase**.
- Fixed **critical defects**, resulting in significantly **more stable application performance**.

SKILLS

Programming & Dev: Python, R, SQL, C#, C++, MATLAB, Flask, Streamlit
ML & AI: PyTorch, TensorFlow, Scikit-learn, Transformers, HuggingFace, LLM, NLP, OpenCV, Time Series Forecasting
Data Engineering & Cloud: Hadoop, PySpark, Apache Beam, Airflow, dbt, Docker, Google Cloud (BigQuery, Dataflow), AWS
Visualization: Tableau, Power BI, SAS Visual Analytics, Matplotlib, Seaborn
Tools & Collaboration: Jira, Confluence, Github
Soft Skills: End-to-End Project Ownership, Research & Benchmarking Mindset, Data Storytelling, Agile Teamwork, Cross-disciplinary Thinking, Adaptability
Languages: English (C1), German (A2 – Actively Improving), Tamil (Native)

KEY ACHIEVEMENTS

On the Spot Award – Tata Consultancy Services: Recognised for resolving a critical Store Finder feature crash, ensuring application stability under production conditions.

On the Spot Award *(Team)* – Tata Consultancy Services: Awarded for successfully delivering the Quotes integration into the checkout feature, enabling a key new business workflow.

Best Team Award – Tata Consultancy Services: Honoured for leading the full migration of a mobile application from .NET to MAUI, modernising the codebase and improving long-term maintainability.

PUBLICATION

Navigation of a Novel Wheelchair for Multi-Level Surfaces – IEEE Conference 2024, Bangalore.
Proposed an autonomous vision-guided wheelchair using deep learning for train compartment detection & path planning.